Preventing back surgery with targeted outreach

Who to contact, with which program, and how we'll know it worked.

Vi Labs · Prepared for Meridian Health

Targeted outreach can avoid expensive back surgeries

The problem

694 lumbar spine surgeries in the past year among Meridian's covered patients: about $13.9M at ~$20,000 each. Under its value-based contract, Meridian pays the full cost.

What's been tried

Outreach for PT and weight management reached 2,189–3,183 patients per wave, a median 288 days after their first back-pain visit. It was never measured: there was no holdout.

The opportunity

Conservative care can prevent some surgeries. Reach the patients most likely to need surgery before they see a surgeon, offer the right program, and measure what it does.

What Meridian asked for: a ranked list for the next wave, a program for each patient, how far down the list to go, and how we'll know it worked.

The expected savings of an outreach must beat its cost

DECISION RULE

$$P(\text{surgery}) \times \text{enrollment lift} \times \text{surgery reduction} \times \text{surgery cost} > \text{outreach cost}$$

Two patients: one worth an outreach, one not

Patient A: recent MRI, opioid prescriptions

Chance of surgery in 12 months	5%
× enrollment lift	11.5 points
× surgery reduction	30%
× surgery cost	$20,000
= expected savings	**$34.50**
Outreach cost	$10.00

Reach out

$34.50 > $10: savings beat the cost.

Patient B: one back-pain visit, nothing since

Chance of surgery in 12 months	1%
× enrollment lift	11.5 points
× surgery reduction	30%
× surgery cost	$20,000
= expected savings	**$6.90**
Outreach cost	$10.00

Skip

$6.90 < $10: savings fall short of the cost.

Hypothetical patients. Each 1% of surgery risk is worth $6.90, so the line falls at 1.45%.

The plan

01 **Who is eligible**

Covered, on a path to surgery, not yet operated on, and reachable.

02 **How the list is ranked**

A 12-month surgery-risk model, tested on a quarter it never saw.

03 **Which program**

Weight management or physical therapy, by BMI and obesity diagnosis.

04 **How many to contact**

Contact while expected savings beat the cost of a contact.

05 **How to measure**

What past outreach can and can't tell us, and how each wave will measure it.

06 **Assumptions**

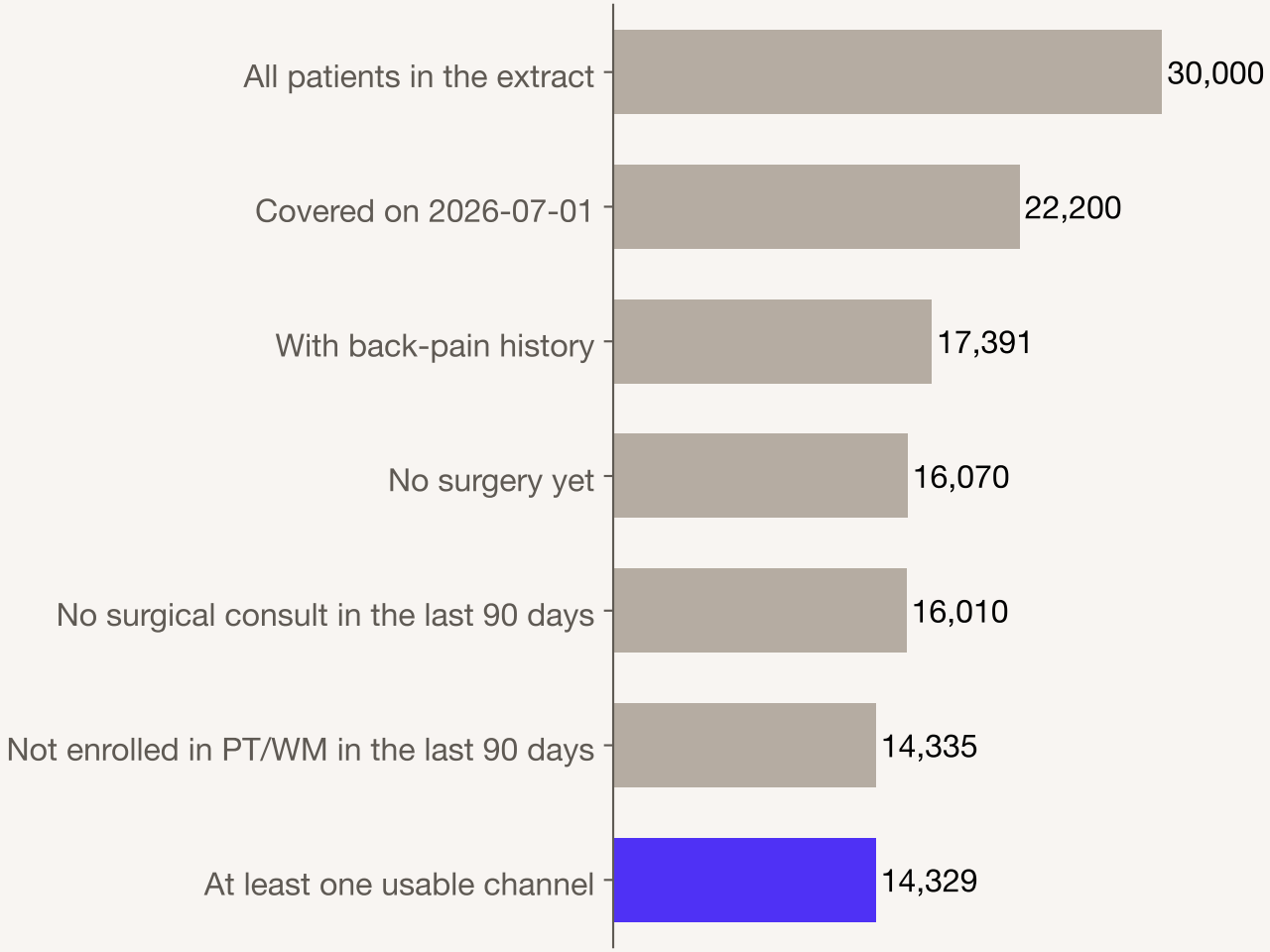
What the answer rests on, and what Meridian should confirm.

01

Who is eligible

Covered, on a path to surgery, not yet operated on, and reachable.

From 30,000 patients to 14,329 we can reach



Everyone in Meridian's data, current members or not.

Still a Meridian member on the day the list is built.

Has seen a clinician for low back pain at least once.

Hasn't had lumbar spine surgery yet, so there is still one to prevent.

Hasn't seen a surgeon recently, when surgery is likely already planned. Older consults stay in.

Isn't already part-way through PT or weight management. Earlier enrollees stay in.

Has a phone, text or email we may use: no opt-outs, invalid numbers or bounces.

02

How the list is ranked

A 12-month surgery-risk model, tested on a quarter it never saw.

What the model predicts

Risk = P(lumbar spine surgery within 12 months, while covered | signals before the list date)

The model: logistic regression

Each signal gets a weight learned from past patients; the weighted sum becomes a probability. Fit on 80,580 quarterly patient snapshots, 2024-07 to 2025-07.

The target: surgery within 12 months, while covered

1 if lumbar spine surgery (EHR or paid claim) in the 12 months after the list date, before leaving Meridian; otherwise 0. Meridian only pays while a patient is covered.

The signals: 36, from the 12 months before the list date

Back-pain care (visits, surgical consults, imaging, injections, PT), spine diagnoses, BMI and obesity, chronic conditions, age and sex. Each one is defined on the next slide.

Rank	12-month risk
1 (highest)	14.5%
1,368 (the cut-off)	1.45%
2,468 (end of band)	1.15%
Average patient	0.85%

Rank 1 is 17× the average patient's risk.

Two patients through the model

Patient A: higher risk

- 58, female, BMI 36 with an obesity diagnosis
- Saw a spine surgeon 6 months ago
- 4 back-pain visits this year, 2 in the last 3 months
- Lumbar MRI; sciatica diagnosis
- First back-pain visit 8 months ago

7.1%

12-month surgery risk; would rank #19 of 14,329.

Above the 1.45% break-even: on the list.

Patient B: lower risk

- 72, male, BMI 23
- Never saw a spine surgeon
- 1 back-pain visit this year, 8 months ago
- No imaging, injections or spine diagnoses
- First back-pain visit over a year ago

0.3%

12-month surgery risk; would rank #13,814 of 14,329.

Below the 1.45% break-even: not contacted.

Hypothetical patients scored by the trained model. Signals not listed are absent. Average risk across the 14,329 ranked patients: 0.9%.

What it looks at: 36 signals from the past 12 months

Counts are log-scaled; every signal covers the 12 months before the list date unless noted.

BACK PAIN AND CARE USE · 19

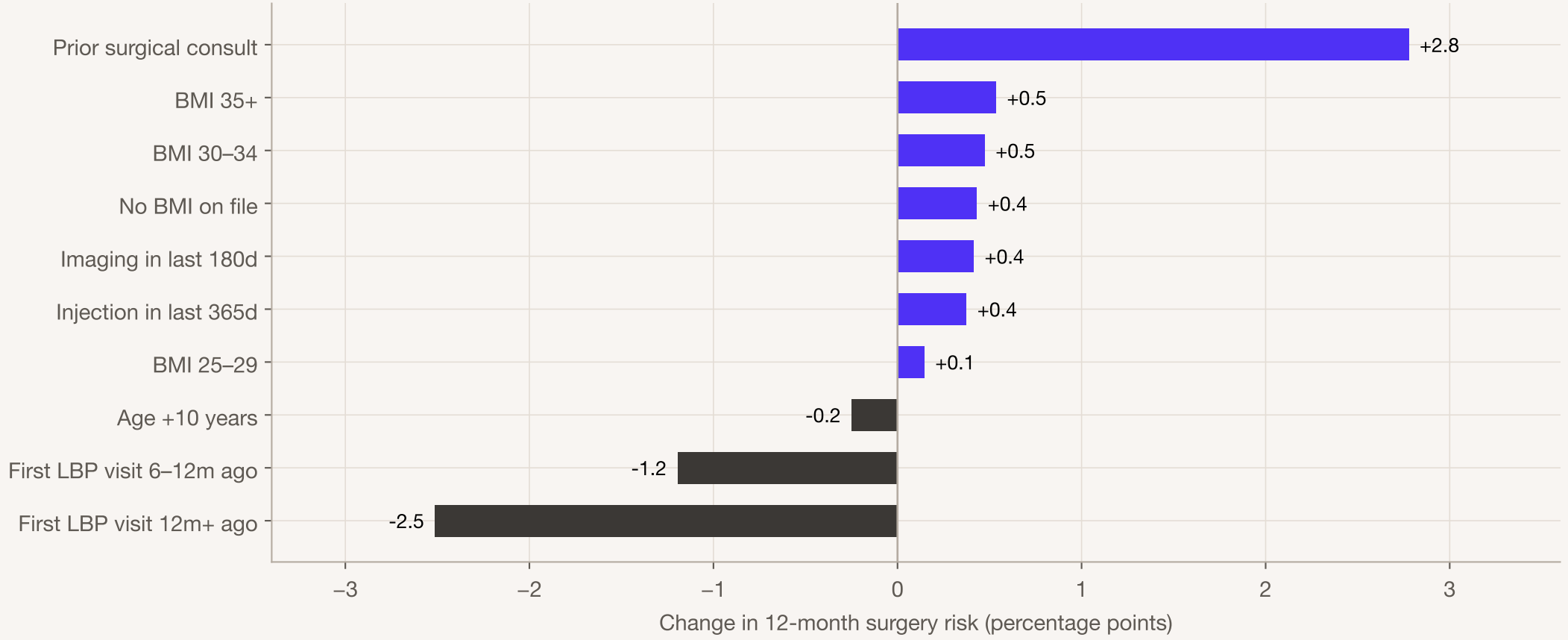
Signal	What it means
Prior surgical consult	Saw a spine surgeon 90–365 days ago
Back-pain visits, 90 days	Low back pain visits in the last 3 months
Back-pain visits, 12 months	Low back pain visits in the last year
Time since last back-pain visit	Years since the latest visit, capped at 2
First back-pain visit 6–12m ago	Vs. a first visit under 6 months ago
First back-pain visit 12m+ ago	Vs. a first visit under 6 months ago
Imaging, 180 days	Any spine imaging in the last 6 months
Lumbar MRI, 12 months	A lumbar spine MRI in the last year
Spinal injection, 12 months	An epidural or other spine injection
PT referral, 12 months	Referred to physical therapy
PT visits	Physical therapy sessions billed
Ortho / pain visits	Visits to orthopedics or pain management
ED visits	Emergency department visits, any cause
Prescription fills	Pharmacy fills, any drug
Earlier PT / WM enrollment	Enrolled 90–365 days ago
Radiculopathy	Nerve pain down the leg (sciatica)
Spinal stenosis	Narrowed spinal canal
Spondylolisthesis	A vertebra slipped out of place
Disc degeneration	Worn lumbar discs

WHO THE PATIENT IS · 17

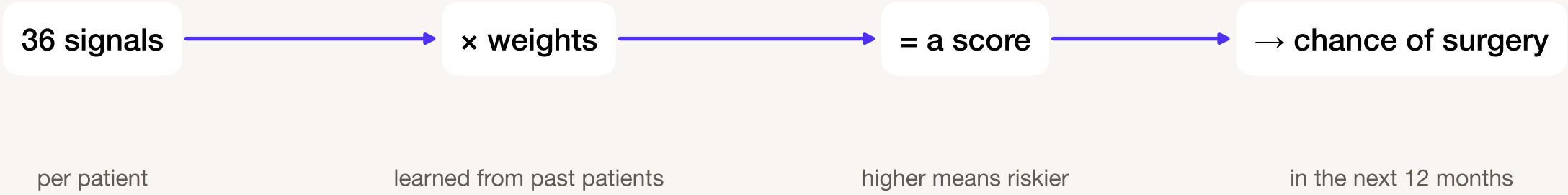
Signal	What it means
Age	Years, at the list date
Under 65	Younger than Medicare age
Male	Vs. female
Sex unknown	Sex not recorded
BMI 25–29	Overweight, vs. BMI under 25
BMI 30–34	Obese, vs. BMI under 25
BMI 35+	Severely obese, vs. BMI under 25
No BMI on file	No BMI code in the last year
Obesity diagnosis	Coded obesity (E66)
Diabetes	Type 2 diabetes
Hypertension	High blood pressure
Heart disease	Coronary artery disease
COPD	Chronic lung disease
Kidney disease	Chronic kidney disease, stage 3
Depression / anxiety	Either diagnosis
Smoker	Nicotine dependence, cigarettes
Osteoporosis	Low bone density

Biggest drivers of risk

Change in a patient's 12-month surgery risk when the signal is present.
Associations, not causes. Outreach history is deliberately not a signal.



How it works: a weighted checklist



- Logistic regression: the score is the log-odds of surgery, so chance = 1 / (1 + exp(–score)).
- Trained on 80,580 patient snapshots, taken quarterly from 2024-07 to 2025-07.

Better than chance: 3.3× a random list

0.73

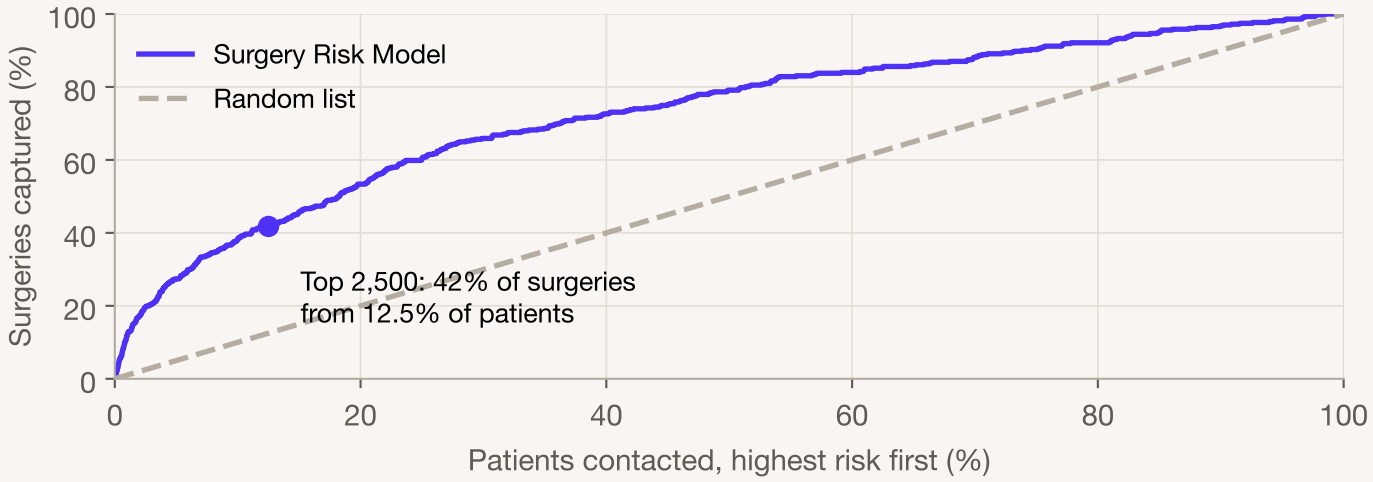
AUC, against 0.50 for random ranking

42%

of surgeries caught in the top 2,500, against 12.5% at random

7.2%

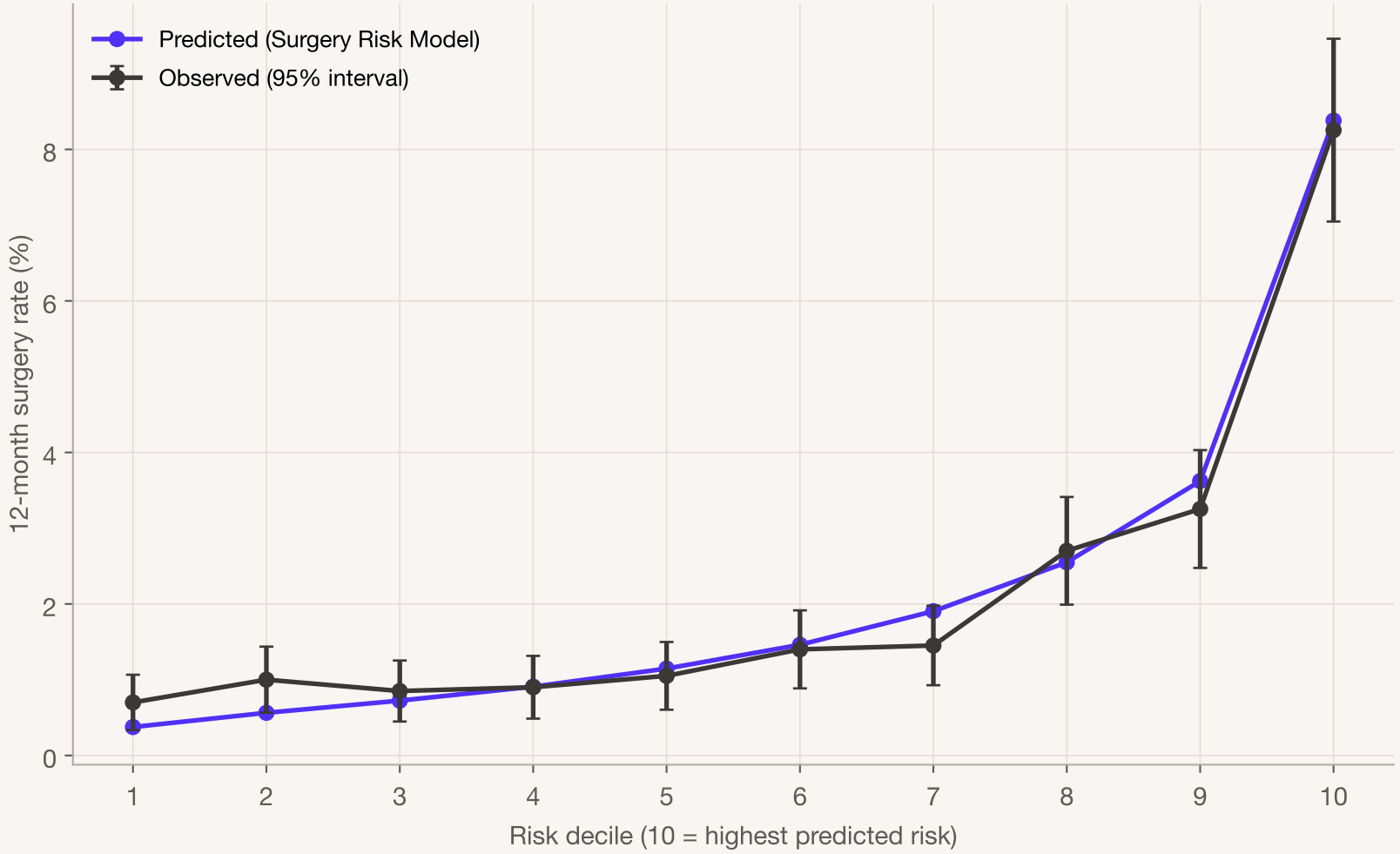
surgery rate on that list, against 2.2% overall



Refit on 2024-07 to 2025-04, tested on the 19,986 patients of 2025-07: a quarter it never saw.

The 2,500 is only the evaluation list size, about the size of past waves. How many to contact is section 4.

Predicted 2.16%, observed 2.16%



Each point is a tenth of the test patients, sorted by risk.

Top decile: 8.4% predicted, 8.3% observed.

Why it matters: the cut-off in section 4 uses the actual probabilities, not just the order.

What the list looks like

Rank	Patient	Risk	Program	Channels	Prior consult
1	eb1284ce	14.5%	physical therapy	SMS, Email	Yes
2	8f5092e7	11.5%	weight management	Phone, SMS	Yes
3	5bac5938	10.3%	physical therapy	Phone, SMS, Email	Yes
6	5088064a	9.0%	physical therapy	Phone, SMS, Email	Yes
7	e389cf26	8.6%	physical therapy	Phone, SMS, Email	Yes
8	ccf99428	8.5%	weight management	Phone, Email	Yes
9	3a7afa0d	8.1%	weight management	Phone, SMS, Email	Yes
10	e95c45ee	7.6%	physical therapy	Phone, Email	Yes
11	a267210c	7.5%	physical therapy	Phone, SMS, Email	Yes
12	d477e2b1	7.4%	physical therapy	Phone, SMS, Email	Yes

The first 10 rows of out/targets.csv. Ranks skip patients randomized to the holdout. Patient tokens are shortened.

03

Which program

Weight management or physical therapy, by BMI and obesity diagnosis.

Weight management where obesity is a factor

Rule: BMI ≥ 30 or an obesity diagnosis → weight management.

Everyone else, including patients with no BMI on file → physical therapy.

Tier	Weight management	Physical therapy
1	409	959
2	310	790

What we can't yet tell you

Two-thirds of patients have no BMI on file and default to PT. History can't pick the program per patient; the pilot could randomize PT vs weight management among patients eligible for both.

BMI 30+ carries about twice the surgery risk of BMI under 25.

04

How many to contact

Contact while expected savings beat the cost of a contact.

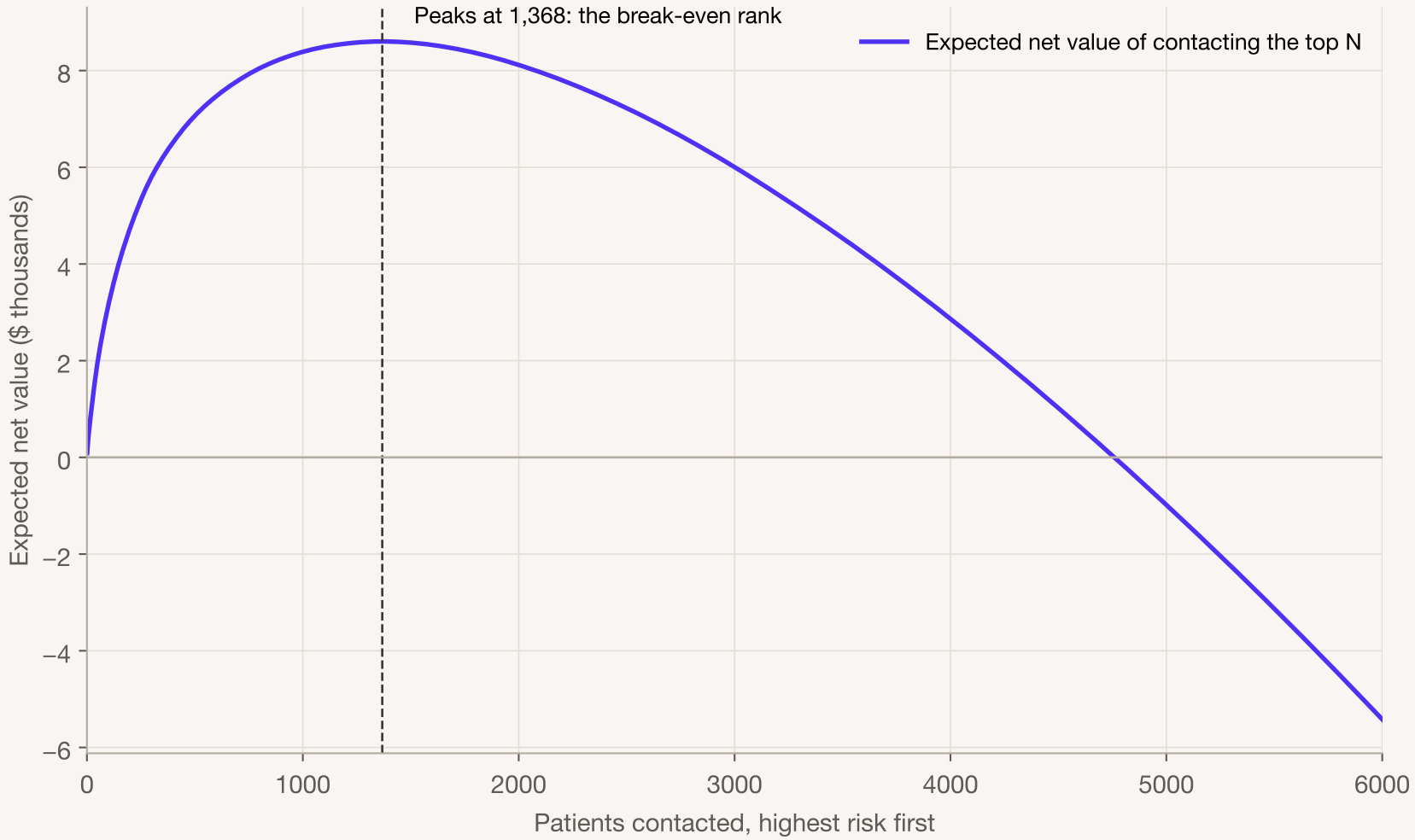
The rule: savings must beat the cost of an outreach

P(surgery) × enrollment lift × surgery reduction × surgery cost > outreach cost

Input	Value	Source
Enrollment lift	11.5%	Estimated from past campaigns (doubly robust, 89% interval 10.6–12.4%)
Surgery reduction per enrollment	30%	Literature assumption: the data can't test it
Surgery cost	$20,000	Medicare 90-day episode, fusion/decompression blend
Outreach cost	$10	Vi call: amortized cost per patient, same for every channel ($5–$10)
Break-even risk	1.45%	$10 ÷ (0.115 × 0.3 × $20,000)

Break-even: contact anyone whose 12-month surgery risk is above 1.45%. Above it, an outreach saves more than its $10 cost; below it, less.

The list is worth most at 1,368 patients



Tier-1 contacts are expected to prevent about 0.9 surgeries this wave: $6.8k net of costs.

That's thin for one wave. The value compounds with repeated waves, and depends on Meridian's real costs and the surgery reduction (section 6).

Contact 1,368, test 1,100, hold out a share

Tier	Who	Contact / hold out	Contacted	Held out	Weight mgmt	Prior consult
1	Above break-even (1,368)	80% / 20%	1,094	274	409	340
2	Learning band (1,100)	50% / 50%	550	550	310	10
—	Below rank 2,468	Not contacted this wave				

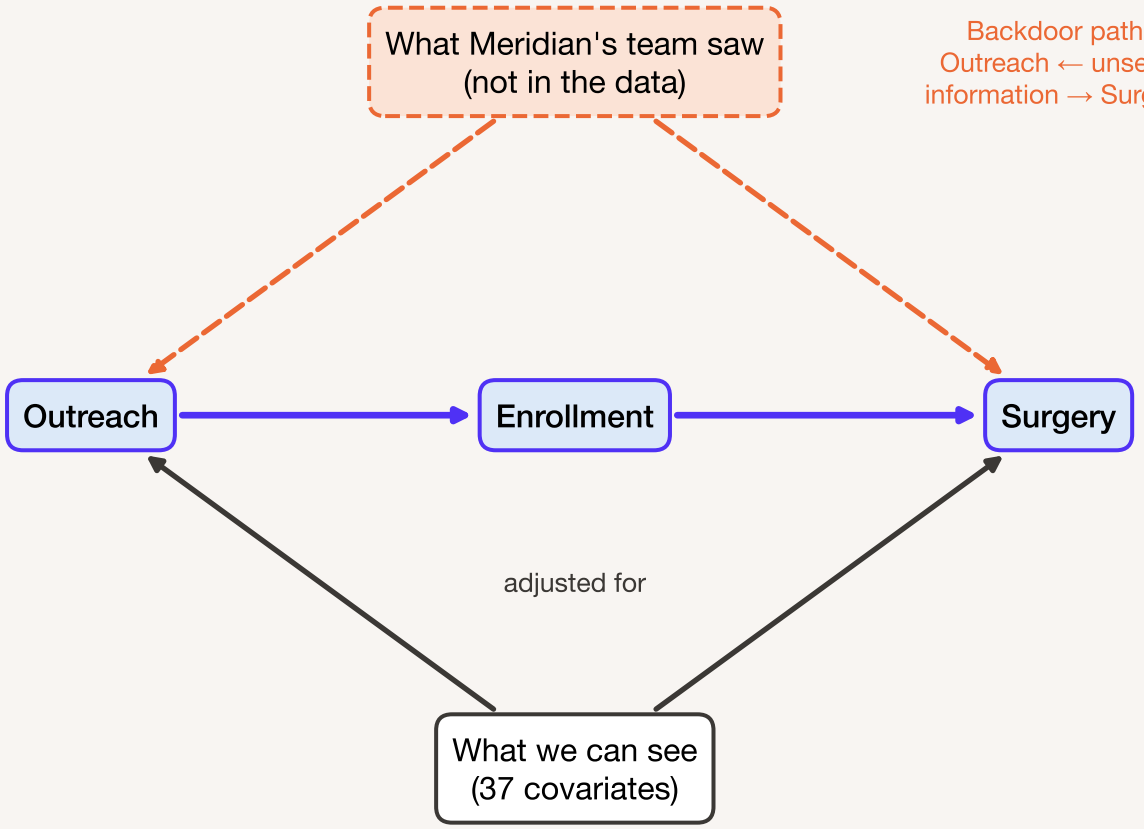
- Randomized within tier and program, with a fixed seed: the assignment is in out/ranking.csv (arm).
- The learning band costs about $0.6k in expected value: the price of checking the cut-off.
- 1,644 patients to contact ≈ 5,710 contact attempts at past rates (3.5 per patient).

05

How to measure

What past outreach can and can't tell us, and how each wave will measure it.

Why history can't settle the surgery question



The raw comparison misleads

Contacted patients had surgery 3.5% of the time vs 2.7% for the rest. At face value, outreach causes surgery.

A negative control flags the bias

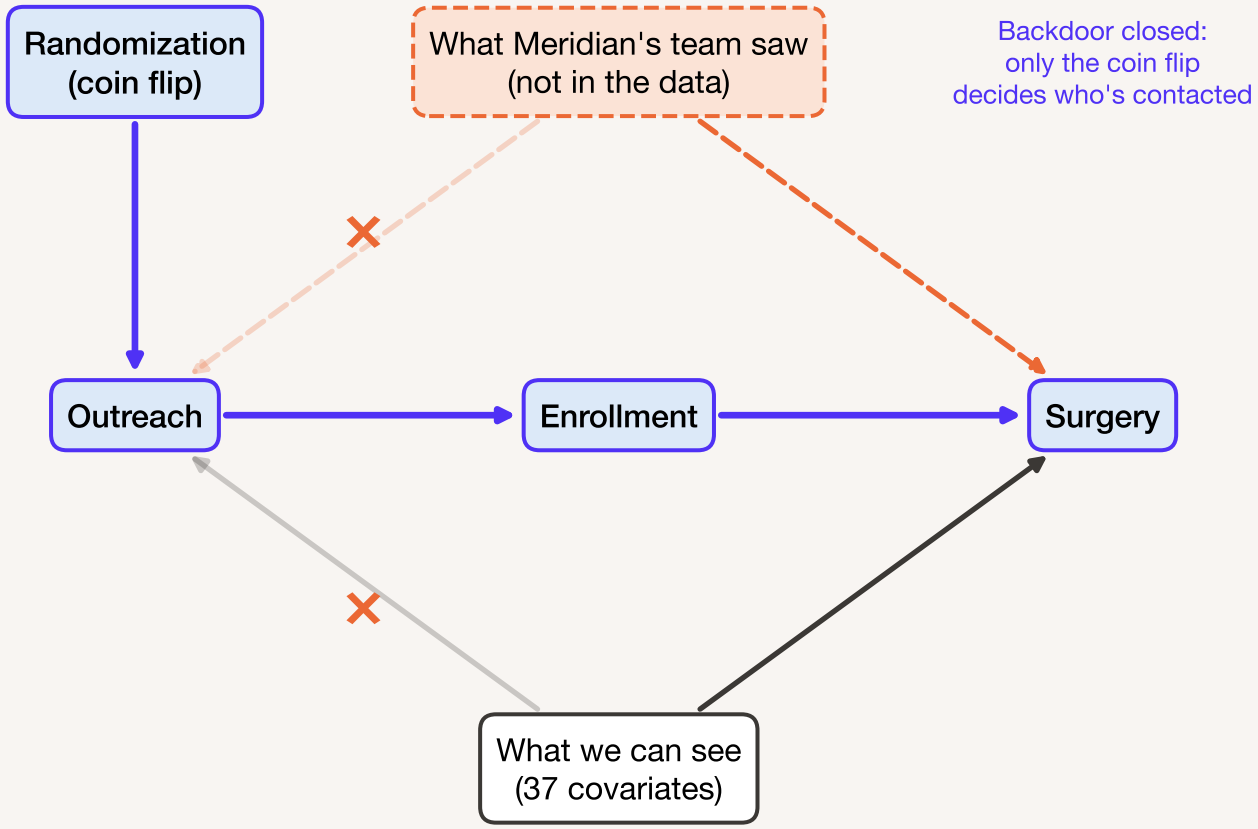
They also had more surgeries in the first 30 days (0.58% vs 0.35%), before any program could act. They were already in the pipeline.

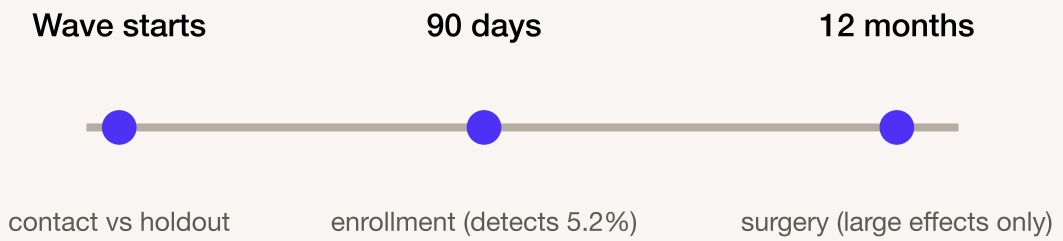
Adjusting isn't enough

After 37 covariates and the negative-control correction: +0.05 points [-0.42, +0.61]. Indistinguishable from zero, and wider than any realistic effect.

Confounder: something that drives both who got contacted and who had surgery. We can estimate the enrollment lift (+11.5%); not the surgery effect.

A random holdout closes the backdoor





- A coin flip decides who's contacted, so nothing else can explain the difference.
- Historical lift: 11.5%; the pilot detects 5.2%.
- A 0.3-point surgery drop needs about 34,000 patients per arm: the surgery readout can only rule out large effects.
- The one ask: Meridian's own campaigns skip holdout patients for 12 months.

Measure every wave the same way

Question	Compare	Read out	One wave can detect
Does contact raise enrollment?	Contacted vs held out, all tiers	90 days	5.2% lift (historical: 11.5%)
Is the cut-off in the right place?	Lift in the learning band vs tier 1	90 days	8.2% lift in tier 1 alone
Does it prevent surgery?	Surgery while covered, contacted vs held out	12 months	Large effects only: pool waves
Which program works better?	PT vs weight management, randomized if eligible for both	90 days, 12 months	Optional next step

Keep the holdout clean

Meridian's own campaigns skip holdout patients for 12 months. Compare by assignment, not by who answered.

Randomize every wave

Within tier and program, with a fixed seed, recorded in out/ranking.csv. A new holdout each wave keeps evidence building.

Feed results back

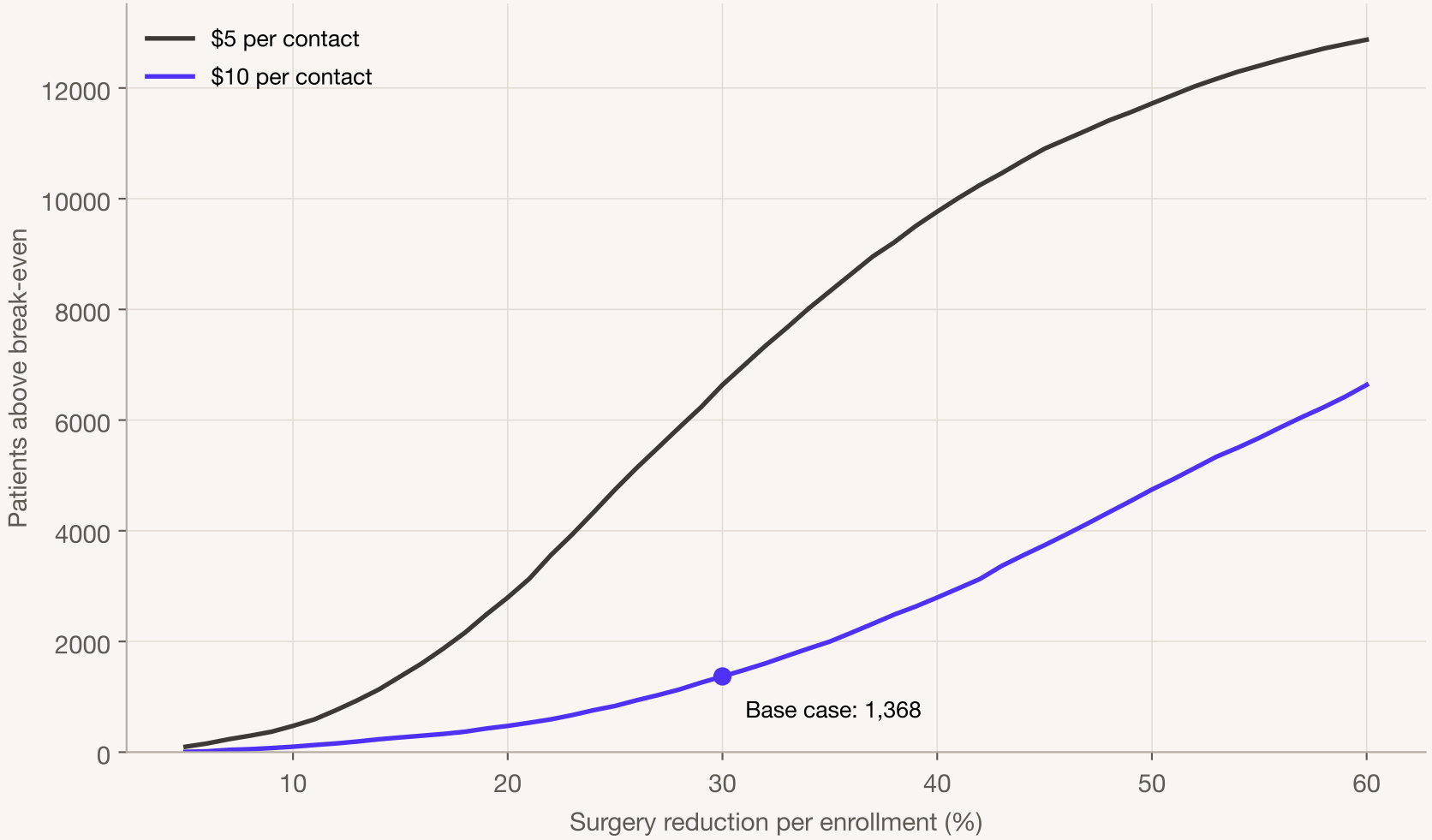
Replace the assumed enrollment lift and surgery reduction in assumptions.py with measured values, then re-score.

06

Assumptions

What the answer rests on, and what Meridian should confirm.

One assumption moves the list most



Patients above break-even as the surgery reduction varies.

It's the number section 5 showed history can't measure.

Across every assumption range, the list runs from 98 to 6,636 patients. The rule stays; only the cut-off moves.

Every assumption, with its source

Assumption	Value	How solid	Source
Enrollment lift from contact	11.5%	Measured	Past campaigns, doubly robust (89%: 10.6–12.4%)
Surgery reduction per enrollment	30%	Assumed	Literature; history can't test it (section 5)
Surgery cost	$20,000	Published	Medicare 90-day episode blend; to confirm
Outreach cost	$10	Vi call	Amortized per patient, same for every channel ($5–$10)
Late-stage window	90 days	Judgment	A consult this recent means already late-stage
Recent-enrollment exclusion	90 days	Agreed	Probably mid-course
Prior-consult patients	Kept, labeled	Agreed	High value for cost; Meridian can filter
Weight-management rule	BMI ≥ 30 or E66	Clinical	Obesity cut-off
Holdout: tier 1 / learning band	20% / 50%	Agreed	Meridian accepts a holdout
Learning band size	1,100	Design	Patients just below break-even, randomized
Capacity	1,644 patients	To confirm	≈ 5,710 contact attempts at past rates

All live in assignment/assumptions.py: change one, re-run the scoring step, and the list updates in seconds.

Meridian to confirm: capacity, outreach cost, contracted surgery cost.